

Your LAST Name _____

Your FIRST Name _____

Your Collaborators

• Name: _____ SID: _____

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Policy Statement

- We encourage you to collaborate, but only in a group of up to *five* current EE 120 students.
- Be sure to write the names of your collaborators at the appropriate location on the next page. Each collaborator must submit a distinct, self-prepared solution document containing original contributions to the collaborative effort.
- To submit your work, you have several options:
 - (a) Print out this homework document and write your solutions on the printout. Then upload a *PDF file* of the scan.
 - (b) Write your solutions on this PDF file using a tablet device. Then upload a *PDF file*.
 - (c) Use your own sheets of paper, but demarcate clearly the space for each problem, part, and subpart. Then upload a *PDF file*.
- Please write neatly and legibly, because *if we can't read it, we can't evaluate it*. Remember that, even though the problem sets are self-graded, we will evaluate your work whenever we check your self grades.
- Unless we explicitly state otherwise, you may expect full credit *only if* you explain your work succinctly, but clearly and convincingly.
- If you are asked to provide a “sketch,” it refers to a well-labeled *hand-drawn* sketch—not a plot generated by a computing device.
- On occasion, a problem set contains one or more problems designated as “optional.” Your solutions to such problems will NOT be evaluated. Nevertheless, you’re responsible for learning the subject matter within their scope.
- If you have a confirmed disability that precludes you from complying fully with these instructions or with any other parameter associated with this problem set, please alert us immediately about reasonable accommodations afforded to you by the DSP Office on campus.

Overview

This problem set explores

- Periodicity
- Casualty and BIBO stability
- LCCDE / DAG representations of systems

HW4.1 (Discrete-Time Periodic Signals) Let $x[n]$ be a discrete-time signal, and let

$$y_1[n] = x[3n] \text{ and } y_2[n] = \begin{cases} x[n/3] & \text{if } n \bmod 3 = 0 \\ 0 & \text{if } n \bmod 3 \neq 0 \end{cases}$$

The signals $y_1[n]$ and $y_2[n]$ respectively represent in upsampled (sped up) and downsampled (slowed down) versions of $x[n]$. For each of the following statements, determine whether it is true. If so, provide a proof. If not, provide a counterexample.

- (1) If $x[n]$ is periodic, then $y_1[n]$ is periodic.
- (2) If $y_1[n]$ is periodic, then $x[n]$ is periodic.
- (3) If $x[n]$ is periodic, then $y_2[n]$ is periodic.
- (4) If $y_2[n]$ is periodic, then $x[n]$ is periodic.

- (1) This statement is true. Suppose $x[n]$ has some fundamental period T . Then, $x[n]$ would have period $3T$. Since $y_1[n]$ samples values at every 3 timesteps from $x[n]$, after T timesteps in $y_1[n]$, it would have traversed $3T$ timesteps from $x[n]$, completing a period. Thus, $y_1[n]$ would also have a period of T . Formally,

$$y_1[n + T] = x[3(n + T)] = x[3n + 3T] = x[3n] = y_1[n]$$

- (2) This statement is false. To provide a counterexample, let

$$x[n] = \begin{cases} 1 & \text{if } n \bmod 3 = 0 \\ n & \text{if } n \bmod 3 \neq 0 \end{cases}$$

Then, $y_1[n] = 1$ since it samples from $x[n]$ at every 3 timesteps. But $x[n]$ is not periodic since it grows unboundedly as n grows large. Intuitively, since $y_1[n]$ is sampled from $x[n]$, we cannot say for sure how $x[n]$ behaves in between those sampled values, therefore we cannot be certain whether it will be periodic too.

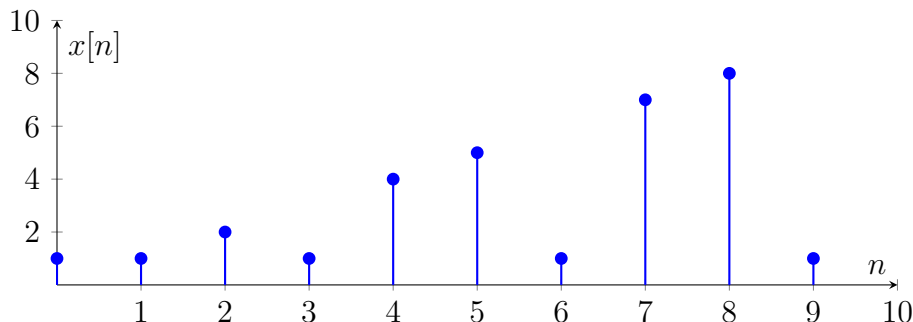


Figure 1: Plot of $x[n]$.

- (3) This statement is true. Suppose $x[n]$ had some period T . Then, $y_2[n]$ would have period $3T$.

If $n \bmod 3 = 0$,

$$y_2[n + 3T] = x\left[\frac{n + 3T}{3}\right] = x\left[\frac{n}{3} + T\right] = x[n/3] = y_2[n]$$

If $n \bmod 3 \neq 0$,

$$y_2[n + 3T] = 0 = y_2[n] \text{ since } (n + 3T) \bmod 3 = n \bmod 3 = 0$$

Intuitively, after every $3T$ timesteps in $y_2[n]$, it would have gone through T timesteps in $x[n]$, which would be a single period. Therefore, $y_2[n]$ would have a period of $3T$.

- (4) This statement is true. Similar to (1), suppose $y_2[n]$ had some period T . Then, it would also have a period of $3T$. For every $3T$ timesteps in $y_2[n]$, $x[n]$ would have traversed T timesteps, meaning that $x[n]$ would also have a period of T . Formally,

$$x[n + T] = x\left[\frac{3n + 3T}{3}\right] = y_2[3n + 3T] = y_2[3n] = x[3n/3] = x[n]$$

HW4.2 (System Properties) Consider each of the following systems independently. For each system, x denotes the input, and y the corresponding output.¹

$$F_1 : [\mathbb{Z} \rightarrow \mathbb{R}] \rightarrow [\mathbb{Z} \rightarrow \mathbb{R}]; \forall n \in \mathbb{Z}, y[n] = x[-n].$$

$$F_2 : [\mathbb{Z} \rightarrow \mathbb{R}] \rightarrow [\mathbb{Z} \rightarrow \mathbb{R}]; \forall n \in \mathbb{Z}, y[n] = x[2n].$$

$$F_3 : [\mathbb{Z} \rightarrow \mathbb{R}] \rightarrow [\mathbb{Z} \rightarrow \mathbb{R}]; \forall n \in \mathbb{Z}, y[n] = \begin{cases} 0 & \text{if } n \text{ is an odd integer} \\ 1 & \text{if } n \text{ is an even integer.} \end{cases}$$

$$F_4 : [\mathbb{Z} \rightarrow \mathbb{R}] \rightarrow [\mathbb{Z} \rightarrow \mathbb{R}]; \forall n \in \mathbb{Z}, y[n] = \begin{cases} x \left[\frac{n}{N} \right] & n \bmod N = 0 \\ 0 & \text{otherwise.} \end{cases}$$

The system F_{10} is called an N -fold upsampler.

Provide a brief but convincing responses to the questions below.

(a) For each system, select the strongest true assertion from the list below.

- (i) The system must be causal.
- (ii) The system could be causal, but does not have to be.
- (iii) The system cannot be causal.

(b) For each system, select the strongest true assertion from the list below.

- (i) The system must be BIBO stable.
- (ii) The system could be BIBO stable, but does not have to be.
- (iii) The system cannot be BIBO stable.

¹The definition of system F_{10} uses the mod function. For any $n \in \mathbb{Z}$ and $N \in \mathbb{N}$, the expression " $n \bmod N$ "—read " n modulo N "—is the unique integer remainder k ($0 \leq k \leq N - 1$) when n is divided by N , e.g., $5 \bmod 3 = 2$ and $-4 \bmod 3 = 2$. Note that N divides $n - k$.

(a) (Causality)

F_1 : **Cannot be causal.** For example, $y[-1] = x[1]$. The output at $n = -1$ depends on the input at $n = 1$.

F_2 : **Cannot be causal.** Let $n = 1$: then, $y[1] = x[2]$. So, the system peeks ahead in time and is not causal.

F_3 : **Must be causal.** Any two arbitrary input signals will produce identical output signals. So, for any inputs $x_1[n]$, $x_2[n]$ such that $x_1[n] = x_2[n]$, $\forall n < N$, we have $y_1[n] = y_2[n]$, $\forall n < N$, as $y_1[n] = y_2[n]$, $\forall n$.

F_4 : **Causal if $N = 1$, otherwise not causal.** If $N = 1$, then $y[n] = x[n]$, which is a causal system.

Otherwise, consider $n = -N$. $y[-N] = x\left[\frac{-N}{N}\right] = x[-1]$, so the system peeks ahead in time and cannot be causal.

(b) (BIBO Stability)

F_1 : **Must be BIBO stable.** Notice that each index of the output signal $y[n]$ is equal to the value of the input signal at a single point in time. In other words, if we were to feed the bounded signal $x[n]$ to the system F_1 , such that $|x[n]| \leq B \in \mathbb{R}$ for all n , then we are assured that the output signal $y[n]$ is also bounded by B .

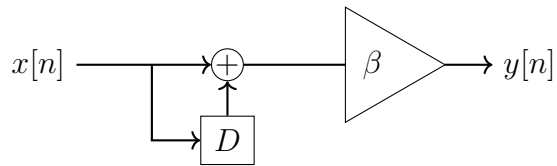
F_2 : **Must be BIBO stable,** by the same reasoning as for system F_1 .

F_3 : **Must be BIBO stable.** The output signal is either 1 or 0, so it is bounded.

F_4 : **Must be BIBO stable,** by the same reasoning as F_1 .

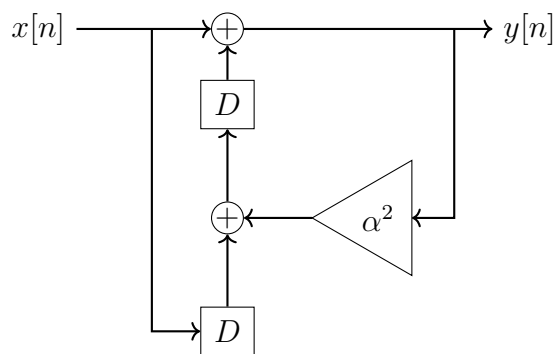
HW4.3 For each of the following delay-adder-gain (DAG) diagrams, give the linear, constant coefficient difference equation (LCCDE) which describes the system.

(a)



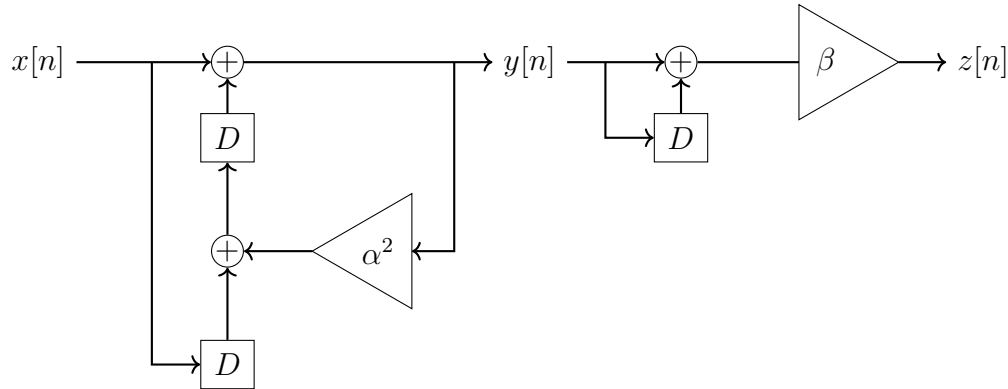
$$y[n] = \beta x[n] + \beta x[n-1]$$

(b)



$$y[n] = x[n] + x[n-2] + \alpha^2 y[n-1]$$

(c)



Note that this is the system from (a) composed with the system from (b). Give an equation in terms of x and z only.

There are many ways to go about this. One way to do this is to convolve the impulse responses from both part (a) and part (b). However, since the convolution is commutative, composing the two systems in either order will create the same system. Thus, we simply have to “plug in” the y value from (a) into (b). This yields

$$z[n] = (\beta x[n] + \beta x[n-1]) + (\beta x[n-2] + \beta x[n-3]) + \alpha^2 z[n-1]$$

Note that doing the substitution the other way doesn’t make sense; substituting x in terms of $y[n] = x[n] + x[n-2] + \alpha^2 y[n-1]$ still leaves you with a y term that is annoying to deal with.

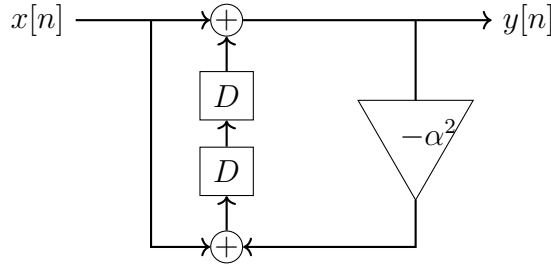
HW4.4 Consider a discrete-time LTI system $F : [\mathbb{Z} \rightarrow \mathbb{R}] \rightarrow [\mathbb{Z} \rightarrow \mathbb{R}]$ whose input and output signals x and y , respectively, satisfy the following linear, constant-coefficient difference equation:

$$y[n] + \alpha^2 y[n-2] = x[n] + x[n-2],$$

where $0 < \alpha < 1$.

- (a) Draw a delay-adder-gain (DAG) block diagram implementation of the system.

We can write $y[n] = x[n] + x[n-2] - \alpha^2 y[n-2]$. At least two delay blocks are needed to implement the system.



- (b) Determine an expression for the frequency response $F : \mathbb{R} \rightarrow \mathbb{C}$ of the system.

By the eigenfunction property, when we let $x[n] = e^{i\omega n}$, then $y[n] = F(\omega)e^{i\omega n}$. Substituting this into the LCCDE,

$$y[n] + \alpha^2 y[n-2] = x[n] + x[n-2]$$

$$F(\omega)e^{i\omega n} + \alpha^2 F(\omega)e^{i\omega(n-2)} = e^{i\omega n} + e^{i\omega(n-2)}$$

$$F(\omega) = \frac{1 + e^{-i2\omega}}{1 + \alpha^2 e^{-i2\omega}}$$

- (c) Suppose $\alpha = 0.95$. Provide a well-labeled sketch of $|F(\omega)|$, the magnitude of the frequency response of the system. Explain why this filter is called a *notch filter*.

Hint: consider what happens at $\omega = \pm \frac{\pi}{2}$ versus elsewhere. Think of $e^{i2\omega}$ as a unit vector rotating in the complex plane, and visualize both the numerator and denominator as vectors. Try sketching this visualization for different values of α .

We have

$$|F(\omega)| = \frac{|1 + e^{-i2\omega}|}{|1 + \alpha^2 e^{-i2\omega}|} = \frac{|e^{i2\omega} + 1|}{|e^{i2\omega} + \alpha^2|}$$

So, we are interested in the ratio of the magnitudes of the two (solid line) vectors shown in Figure 5. Since $\alpha = 0.95$, this ratio is roughly 1 at every

frequency except for $\omega = \pm \frac{\pi}{2}$, when the numerator becomes zero. Therefore, the magnitude of the frequency response is as shown in Figure 6; this filter is called a *notch filter* because it passes all frequencies except for two, where there is a “notch” in the magnitude plot.

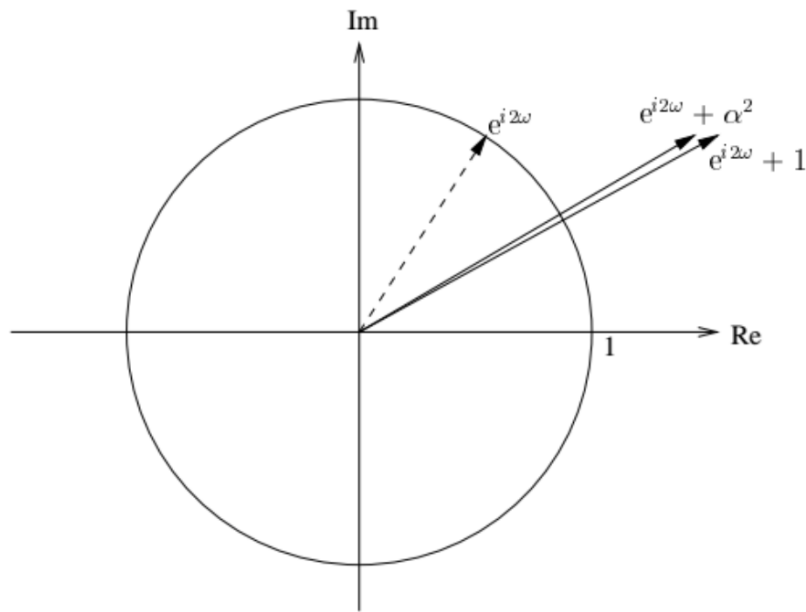


Figure 5: 3.5(c): Geometric analysis

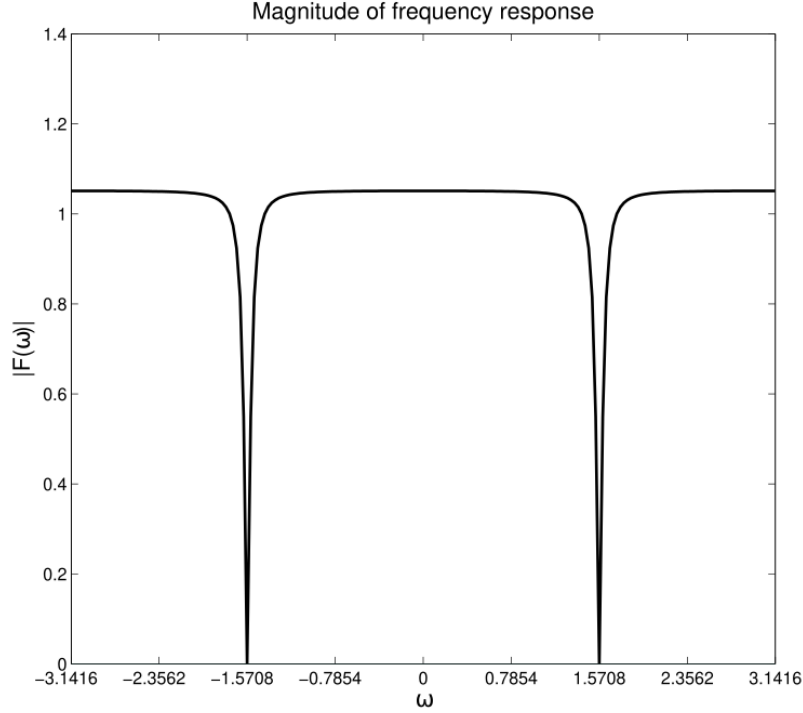


Figure 6: 3.5(c): Notch filter

(d) Suppose $\alpha = 0.95$ and that the input signal is characterized by

$$\forall n \in \mathbb{Z}, \quad x[n] = \cos\left(\frac{\pi}{2}n\right) + \frac{1}{3} \sin\left(\frac{\pi}{4}n\right) + 3 + (-1)^n.$$

Using little to no mathematical manipulation, determine a reasonable approximation for the corresponding output signal values $y[n]$.

What assumption did you have to make about the phase response $\angle F$ that enabled you to approximate the output signal y ? Why was your assumption about the phase reasonable?

As this is a linear system, we can break up the input $x(n)$ into its addends. The discerning student will pick up on an interesting fact – these constituent parts are all sinusoids with frequencies $\omega_1 = \frac{\pi}{2}$, $\omega_2 = \frac{\pi}{4}$, $\omega_3 = 0$, and $\omega_4 = \pi$. We can see from our magnitude response in part (b) above that sinusoids of nearly all frequencies will pass through with no significant change in amplitude. However, it's also clear that ω_1 is right at the sweet spot of the notch filter, and its associated sine wave won't be appearing as an addend in y . Its amplitude drops to zero. If we assume that the phase response is zero (or near zero) for all the other frequencies, we can approximate the output signal:

$$y[n] = \frac{1}{3} \sin\left(\frac{\pi}{4}n\right) + 3 + (-1)^n$$

We can assume the phase response is nearly zero because the vectors in the numerator and denominator are nearly identical and have similar phases.